Morgan Stanley | RESEARCH

UPDATE

October 15, 2025 10:59 PM GMT

Global Macro Commentary | North America

October 15: Balanced Beige Book

MORGAN STANLEY & CO. LLC
Bradley Tian
Strategist
Bradley.Tian@morganstanley.com +1 212 761-0826
Molly Nickolin
Strategist
Molly.Nickolin@morganstanley.com +1 212 761-3592
Lingdi Xu
Economist
Lingdi.Xu@morganstanley.com +1 212 761-2957
Sofia Palacios
Strategist
Sofia.Palacios@morganstanley.com +1 212 761-0428

MORGAN STANLEY ASIA LIMITED+
Gek Teng Khoo
Strategist
Gek.Teng.Khoo@morganstanley.com +852 3963-0303

MORGAN STANLEY MUFG SECURITIES CO., LTD.+
Hiromu Uezato
Strategist
Hiromu.Uezato@morganstanleymufg.com +81 3 6836-8431

MORGAN STANLEY INDIA COMPANY PRIVATE LIMITED+
Nimish M Prabhune
Strategist
Nimish.Prabhune@morganstanley.com +91 22 6996-1862

MORGAN STANLEY & CO. INTERNATIONAL PLC+
Maria Chiara Russo
Strategist
Maria.Chiara.Russo@morganstanley.com +44 20 7677-3499

Risk fluctuates on mixed US-China trade signals, with Bessent floating pause on tariffs & export controls; balanced risks in Fed's Beige Book; strong US Empire manufacturing; US federal layoffs temporarily blocked; low USD/CNY fixing below 7.10; DXY at 98.793 (-0.3%); US 10y at 4.028 (-0.4bp).

Risk sentiment fluctuates throughout the day on US-China trade headlines, but ends the day fairly strong.

Developed Markets

- **Front-end US rates sell off after a strong Empire manufacturing survey and balanced Beige Book trim expectations for future Fed cuts.** The Empire survey rebounds strongly to 10.8 in October from -8.7 in September, a potential sign of strong manufacturing amid the lack of economic hard data. Afterwards, 2y Treasuries sell off 2bp as the market-implied probability of Fed cuts in December, January, and March decrease by ~5% each. Yields remain elevated as the Fed's Beige Book shows "economic activity changed little" last month, with similarly little change in unemployment, but more price pressures than the prior Beige Book as "district reports indicated that input costs increased at a faster pace due to higher import costs."

- **The dollar weakens and US equities rally (S&P 500: +0.4%) as risk sentiment recovers after Treasury Secretary Bessent proposes a longer pause on US tariffs on China in exchange for China delaying its rare earth export controls.** Bessent adds that neither country wants a full economic decoupling, in line with our view that the most likely outcome is another limited agreement that largely recreates the prior détente regarding export controls. However, Bessent later suggests that the World Bank end support for China, and President Trump says the US and China are in a "trade war." Defense stocks lag the equity rally (S&P Aerospace & Defense: -1.6%) after Bessent says the administration may ask defense companies to reduce stock buybacks.

- **As the US government shutdown continues to show little progress, US courts temporarily block the administration's efforts to lay off over 4,000 federal workers,** reducing the risk of larger labor market impacts after officials earlier proposed up to 10,000 layoffs. President Trump orders military employees to be paid despite the shutdown, potentially reducing near-term economic and political pressures. Prediction markets now price a 75% likelihood of the shutdown lasting longer than a month. Meanwhile, Treasury Secretary Bessent confirms prior reporting that five candidates are being considered for the next Fed chair, and he intends to present President Trump with three or four options in December.

- **Japanese rates twist flatten in a partial reversal of Tuesday's steepening after Nikkei reports a possible BoJ rate hike this winter.** 2y JGBs sell off 1bp as pricing for October and December rate hikes slightly recovers. Long-end JGBs rally strongly into the afternoon's 20y auction and continue to rally afterwards given that the auction is digested smoothly, with a cut-off price of 100.20. JPY strengthens 0.5% against USD as rate differentials shift in favor of Japan, while the Nikkei retraces Tuesday's weakness, gaining 1.8%.

- **European duration continues to rally on the back of Tuesday's constructive fiscal developments in France.** German bunds bull-flatten (2y: -1bp; 30y: -5bp), with more muted moves in EUR swaps. French OATs lead a continued tightening in EGB spreads. The CAC 40 rallies 2%, outperforming the Euro Stoxx 50 (+1%) as market sentiment remains positive ahead of tomorrow's no-confidence vote given that Prime Minister Lecornu gained political support following his suspension of the proposed pension reforms.

- **Among G10 currencies, NOK outperforms (NOK/USD: -0.7%) as Norway's 2026 budget comes broadly in line with expectations.** CAD underperforms the dollar weakness, with USD/CAD holding steady as Canadian rates rally 2bp despite the US sell-off and a slower-than-expected decline in manufacturing sales (-1.0% m/m; C: -1.5%). Elsewhere, AUD/USD strengthens 0.4% supported by a low USD/CNY fixing, but still lags the G10 given its exposure to US-China trade tensions.

Emerging Markets

- **AxJ currencies rally as risk appetite rebounds and the USD/CNY fixing breaks below the widely watched 7.10 level.** In China, CNH strengthens 0.2% after the release of the fixing while CNY NDIRS yields remain largely stable amid soft September inflation (-0.3% y/y; C: -0.2%) and credit growth data (RMB1290bn; C: RMB1460bn). KRW rallies 0.4% back to ~1,422 per USD alongside local equity gains (KOSPI: +2.7%), while KRW NDIRS yields remain largely stable despite the announcement of stricter-than-expected property market cooling measures.

- **India leads regional FX gains, rallying 0.8% to 88.07 per USD amid unconfirmed reports of RBI intervention,** with a wider-than-expected September trade deficit (-$32bn; C: -$28bn) putting only a temporary dent in currency gains. Indian rates rally 1-3bp in a bull-flattening move, supported by reports that the RBI is conducting FX swaps to manage liquidity in the system, which stands at a surplus of around INR1.2tr. Meanwhile, Indonesia lags regional peers, with IDR strengthening just 0.2% even as local rates outperform (10y: -5bp).

- **LatAm currencies rebound in line with global risk appetite, while LatAm**

rates outperform US Treasuries. As the US considers doubling aid for Argentina to $40bn, Argentine bonds and equities (MERVAL: +1.5%) rise, but USD/ARS gains 0.4% despite another US purchase of ARS. Most other regional currencies trade stronger against the dollar, with Colombia leading gains (USD/COP: -0.8%). Colombia's IBR curve also sees the most pronounced bull-flattening, with long-end yields falling nearly 10bps as production and sales prints come in softer than expected.

Central Bank Monitor

Developed Markets

- **Fed Governor Miran** highlighted emerging risks to the economic outlook stemming from China's rare earth export controls. He noted that inflation is trending back toward 2%, with substantial disinflation expected within a year. Miran added that trade uncertainty increases the urgency for Fed rate cuts and emphasized that US-China trade tensions pose significant downside risks. He said lower immigration is unlikely to create wage pressures and stressed the need for fair treatment of all policies in inflation analysis.
- **Fed Governor Waller** said AI is expected to boost GDP and national income, though job gains may take a few more years due to the rapid pace of development. He emphasized that disruption is necessary to unlock technological benefits and advocated continued investment in AI, though it primarily enhances productivity.
- **ECB Governing Council Member Nagel** said there are currently no grounds to change interest rates, emphasizing the need for vigilance regarding inflation risks.
- **RBA Governor Bullock** described the RBA's current monetary policy as "marginally tight."
- **RBA Assistant Governor Kent** said that the drag from tight financial conditions is easing, helping the economy stay near full employment with inflation returning to target.
- **RBA Assistant Governor Hunter** said Australia's underlying inflation in the three months through September will likely be stronger than the RBA had anticipated.
- **ECB Governing Council Member Muller** said that inflation risks are more or less balanced, but China's export controls could be inflationary.
- **ECB Governing Council Member Villeroy** reiterated that the ECB's next move is more likely to be a cut than a hike.

Exhibit 1: G4 Rates Closes (5pm NY)

	US			Germany			UK			Japan		
Tenor	15-Oct	1d Δ (bp)	2wk	15-Oct	1d Δ (bp)	2wk	15-Oct	1d Δ (bp)	2wk	15-Oct	1d Δ (bp)	2wk
2y	3.497	1.7		1.919	(1.3)		3.892	(0.3)		0.894	1.1	
3y	3.500	1.1		1.979	(2.5)		3.881	(2.1)		0.980	0.2	
5y	3.616	0.9		2.164	(3.6)		3.999	(3.7)		1.187	1.2	
7y	3.803	0.5		2.305	(3.7)		4.136	(4.2)		1.411	1.1	
10y	4.028	(0.4)		2.570	(3.9)		4.542	(4.8)		1.647	(0.1)	
20y	4.593	(0.1)		3.047	(4.4)		5.230	(5.3)		2.693	(1.5)	
30y	4.625	(0.7)		3.145	(4.6)		5.342	(5.1)		3.165	(4.7)	

Source: Morgan Stanley Research, Bloomberg

Exhibit 2: Inflation Closes (5pm NY)

US TIPS				10y Real Yields, BEI							
Tenor	15-Oct	1d Δ (bp)	2wk	10y	15-Oct	1d Δ (bp)	2wk	10y	15-Oct	1d Δ (bp)	2wk
5y	1.231	1.6		DBRi	0.618	(2.8)		BTPi	1.686	(1.9)	
BEI	2.356	(0.9)		BEI	1.741	(1.0)		BEI	1.715	(2.1)	
10y	1.725	0.2		UKTi	1.564	(2.7)					
BEI	2.300	(0.7)		BEI	3.049	(2.3)					
30y	2.403	(0.3)		JGBi	-0.001	(1.4)					
BEI	2.223	(0.4)		BEI	1.620	1.6					

Source: Morgan Stanley Research, Bloomberg

Exhibit 3: Macro Closes (5pm NY)

Majors				EM				Equities / Commodities			
Spot	15-Oct	1d Δ (%)	2wk	Spot	15-Oct	1d Δ (%)	2wk	Index	15-Oct	1d Δ (%)	2wk
DXY	98.79	(0.3)		CNH	7.13	(0.1)		S&P	6671.06	0.4	
EUR	1.165	0.3		INR	88.07	(0.8)		Stoxx	5605.03	1.0	
GBP	1.340	0.6		THB	32.55	(0.7)		FTSE	9424.75	(0.3)	
JPY	151.05	(0.5)		KRW	1422.50	(0.4)		Nikkei	47672.67	1.8	
CHF	0.797	(0.5)		ZAR	17.35	(0.2)		VIX	20.64	(0.8)	
AUD	0.651	0.4		MXN	18.46	(0.2)		Gold (S)	4207.48	1.6	
CAD	1.404	(0.0)		BRL	5.45	(0.5)		WTI (Fut)	58.27	(0.7)	

Source: Morgan Stanley Research, Bloomberg

Exhibit 4: G4 Central Bank Implied Pricing

Central Bank OIS															
Fed	Implied	1d Δ (bp)	Total Δ	ECB	Implied	1d Δ (bp)	Total Δ	BOE	Implied	1d Δ (bp)	Total Δ	BOJ	Implied	1d Δ (bp)	Total Δ
EFFR (10/14)	4.100	0.0	--	EONIA (10/14)	1.927	0.0	--	SONIA (10/14)	3.967	0.0	--	TONAR (10/14)	0.477	0.0	--
10/29/25	3.862	(0.0)	(23.8)	10/30/25	1.921	0.2	(0.6)	11/06/25	3.936	0.1	(3.1)	10/30/25	0.511	1.0	3.4
12/10/25	3.630	1.3	(47.0)	12/18/25	1.895	0.6	(3.2)	12/18/25	3.871	(0.1)	(9.7)	12/19/25	0.603	0.7	12.6
01/28/26	3.493	0.5	(60.7)	02/05/26	1.877	1.1	(5.0)	02/05/26	3.728	(1.6)	(23.9)	01/23/26	0.668	1.9	19.1
03/18/26	3.366	1.0	(73.4)	03/19/26	1.828	1.4	(9.9)	03/19/26	3.694	(0.8)	(27.3)				
04/29/26	3.293	1.0	(80.7)	04/30/26	1.804	1.1	(12.3)	04/30/26	3.596	(0.3)	(37.1)				
06/17/26	3.148	0.8	(95.2)	06/11/26	1.771	(0.6)	(15.6)	06/18/26	3.560	(1.6)	(40.7)				

Source: Morgan Stanley Research, Bloomberg

The Day Ahead

Developed Markets

- 7:00am BST - **UK August Monthly GDP** (MSe: 0.1% m/m; C: 0.1%; P: 0.0%)
- 7:00am BST - **UK August Index of Services** (MSe: 0.1% m/m; C: 0.1% m/m; P: 0.1%)
- 7:00am BST - **UK August Industrial Production** (MSe: 0.0% m/m; C: 0.2%; P: -0.9%)
- 2:00pm BST - **ECB Governing Council Member Wunsch** will speak in Washington
- 4:45pm BST - **ECB Chief Economist Lane** will speak in Washington
- 5:00pm BST - **ECB President Lagarde** will speak in Washington
- 8:30am EST - **US October Philadelphia Fed Business Outlook** (C: 10; P: 23.2). We anticipate a moderation in October's Philly Fed Manufacturing Index following September's sharp rebound to +23.2 from -0.3 in August — the highest reading since January. That surge signaled a notable recovery in regional manufacturing activity, making a pullback or normalization in the upcoming release a reasonable expectation
- 8:30am EST - **US October New York Fed Services Business Activity** (P: -19.4). September's headline index fell to -19.4, the weakest reading since April and the seventh consecutive monthly contraction in New York State's service sector. The persistent decline underscores ongoing challenges facing service firms, and we expect this weakness to extend into October

- 10:00am EST - **US October NAHB Housing Market Index** (C: 33; P: 32). Points to ongoing weakness in single-family building
- 9:00am EST - **Fed Governor Barr** and **Fed Governor Miran** will speak
- 10:00am EST - **Fed Vice Chair for Supervision Bowman** will speak at the Stress Testing Research Conference
- 6:00pm EST - **Minneapolis Fed President Kashkari** will speak in South Dakota
- 8:30pm ET - **Australia September Employment Change** (MSe: 30.0K; C: 20.0K; P: -5.4K); **Unemployment Rate** (MSe: 4.2%; C: 4.3%; P: 4.2%) & **Participation Rate** (MSe: 66.8%; C: 66.8%; P: 66.8%)

Emerging Markets

- Thu 11:00am HKT - **Korea September Total Bank Lending to Household** (P: KRW1168.3tr)
- 8:00am EST - **Brazil August Economic Activity** (P: 1.15% y/y; C: 0.70%). Markets expect a deceleration in economic activity, but fiscal concerns remain elevated

Economic Releases

Developed Markets

- **US October Empire Manufacturing:** 10.7 (C: -1.8; P: -8.7). A healthy rebound in October after a month of weakness at -8.7 and +11.9 in the prior two months. On an ISM-adjusted basis, it read 52.7 — above trend and approaching recent highs. Solid sign for October manufacturing, but it's only the first of the regional surveys and can be noisy from month to month.
- **US Fed October 2025 Beige Book:** Overall economic activity remained largely unchanged, mirroring the tone of the September release. However, price pressures intensified, with several districts noting faster input cost increases — particularly from imports — and more firms in manufacturing and retail passing these costs on to consumers.
- **Sweden September Final CPIF-ex Energy:** 2.7% y/y (MSe: 2.7%; C: 2.7%; P: 2.7%); **Final CPIF:** 3.1% y/y (MSe: 3.1%; C: 3.1%; P: 3.1%) & **Final CPI:** 0.9% y/y (MSe: 0.9%; C: 0.9%; P: 0.9%)
- **Euro Area August Industrial Production ex Construction:** -1.2% m/m (MSe: -3.1%; C: -1.8%; P: 0.3%). The decline was milder than expected.

Emerging Markets

- **China September Headline CPI:** -0.3% y/y (MSe: -0.2% y/y; C: -0.2%; P: -0.4%); **Core CPI:** 1.0% y/y (MSe: 0.9% y/y; P: 0.9%) & **PPI:** -2.3% y/y (MSe: -2.2% y/y; C: -2.3%; P: -2.9%)
- **China September New TSF:** RMB3534bn (MSe: RMB3150bn; C: RMB3349bn; P: RMB2567bn); **New CNY Loans:** RMB1290bn (MSe: RMB1450bn; C: RMB1460bn; P: RMB589bn) & **M2 Money Supply:** 8.4% y/y (MSe: 8.5%; C: 8.5%; P: 8.8%)
- **Korea August M2 Money Supply:** 1.3% m/m (P: 0.8%)
- **Indonesia August External Debt:** $431.9bn (P: $430.8bn)

- **Philippines August Overseas Cash Remittances:** 3.2% y/y (C: 2.9%; P: 3.0%)
- **India September Trade Balance:** -$32.2bn (C: -$27.5bn; P: -$26.5bn); **Exports:** 6.7% y/y (P: 6.7%) & **Imports:** 16.7% y/y (P: -10.1%); **Unemployment Rate:** 5.2% (P: 5.1%)
- **Brazil August Retail Sales:** 0.4% y/y (P: 1.0%; C: 0.2%). Retail sales slowed, though the decline was milder than consensus expectations. The government continues to face setbacks in advancing revenue-raising measures and is debating new proposals to offset a projected deficit. Markets are pricing in roughly 4bps of additional cuts for the remainder of 2025, and about 225bps of easing from 2026 through mid-2027, with the terminal rate near 12.62%
- **Colombia August Retail Sales:** 12.4% y/y (P: 17.9%; C: 13.4%). A larger deceleration in retail sales compared to consensus, and fiscal concerns continue to linger. Markets are currently pricing in ~3bps of cuts for the remainder of this year and ~47bps between 2026 and early 2027, with a projected terminal rate near 8.76%

For more details, please refer to our economists' weekly reports and data previews:

- *US*
- *Japan*
- *Europe*
- *Asia/Pacific*
- *LatAm*
- *CEEMEA*

Please click here if you would like to receive our weekly strategic advice on global rates, FX, and EM. See our latest Global Macro Strategist: As the Grip of Tariffs Tightens and recent publications:

- US Rates Strategy: An Introduction to STRIPS
- EM Sovereign Credit Strategy: Rich & Cheap Watch

G4 Smarter (Beta) Trading Partners

MORGAN STANLEY & CO. LLC

Matthew Hornbach, CMT
Matthew.Hornbach@morganstanley.com
+1 212 761-1837

Since 2000, enhancements to a G4 10y government bond futures momentum strategy have produced higher Sharpe ratios and stronger returns relative to total return government bond indices for the G4: US, Germany, Japan, and the UK. See A "Smarter" (Beta) Way to Trade G4 10y Futures Duration? for more information on these strategies.

Exhibit 5: Trading Signals for G4 Smarter (beta) Trading Strategy

Current Risk, G4 10y Futures	G4 Strategy Weight	Trade Longs Portfolio	Fade Shorts Portfolio	Total Risk Trade Longs Only	Total Risk Trade Longs/Fade Shorts (max 200%)	Trade Longs Portfolio Entry Date	Trade Longs Portfolio Exit Date	Fade Shorts Portfolio Entry Date	Fade Shorts Portfolio Exit Date
JB 10y Future	32.50%	100%	100%	100%	200%	10/15/2025	11/20/2025	9/19/2025	10/28/2025
GE 10y Future	29.25%	100%	100%	100%	200%	10/7/2025	11/11/2025	9/15/2025	10/20/2025
US 10y Future	30.50%	100%	100%	100%	200%	10/14/2025	11/19/2025	9/24/2025	10/30/2025
UK 10y Future	7.75%	100%	100%	100%	200%	10/15/2025	11/19/2025	9/25/2025	10/30/2025

Source: Morgan Stanley Research

Trading Strategy 1 – "Trade Longs/Fade Shorts"

When the 5-day moving average crosses above the 20-day moving average, buy the futures contract (long duration) and hold for a 25-business-day period. When the 5-day moving average crosses below the 20-day moving average, buy the futures contract and hold for a 25-business-day period. In short, this strategy buys futures when the Simple Moving Average Crossover (SMAX) generates both a long and a short signal, given the historical outperformance of long signals traded long and the underperformance of short signals traded short. Given that the SMAX could generate both a long and a short signal within the predefined holding period, an investor may have a 200% long position since each of the two signals would be traded in separate portfolio sleeves.

Trading Strategy 2 – Trade "Longs Only"

When the 5-day moving average crosses above the 20-day moving average, buy the futures contract (long duration) and hold for a 25-business-day period. When the 5-day moving average crosses below the 20-day moving average, do nothing. In short, an investor *only* trades long signals initiated by the SMAX given their historical precedent to outperform.

US Holiday Trading Strategy

Currently Active Strategies

None.

Combining the individual trading signals based on 10y Treasury future (TY) price patterns before and after US holidays has generated an annual strategy with no down years since 1987. See Trading 10y Treasury Futures Around US Holidays for more information.

Using historical data since 1983, we created a trading strategy using 10y Treasury futures (TY) that trades 11 times per year, either long or short, before or after the following US holidays: New Year's Day, Martin Luther King Day, Presidents' Day, Good Friday, Memorial Day, Independence Day, Labor Day, Columbus Day, Thanksgiving, and Christmas. Exhibit 6 displays the trading rules:

Exhibit 6: Trading rules for US holiday trading model using US 10y futures contracts (TY)

Holiday	Day of Week	Direction	Before/After Holiday	Trading Rule (business days, 3pm NY unless specified)	Trade Entry (business days)	Trade Exit (business days)
New Year's Day	Rotates	Long TY	After	Buy TY New Years +1, Sell TY New Years +4	New Year + 1	New Year + 4
Martin Luther King Day 1	Monday	Long TY	Before	Buy TY Tuesday before MLK, Sell Friday before MLK	MLK - 4	MLK - 1
Martin Luther King Day 2	Monday	Short TY	After	Sell TY Tuesday after MLK, Buy Friday after MLK	MLK + 1	MLK + 4
President's Day	Monday	Long TY	After	Buy TY Tuesday after Pres. Day, Sell TY Friday after Pres. Day	Pres. Day + 1	Pres. Day + 4
Good Friday	Friday	Long TY	After	Buy TY Monday after Good Friday, Sell TY Friday after Good Friday	Good Friday + 1	Good Friday + 5
Memorial Day	Monday	Long TY	After	Buy TY Tuesday after Mem. Day, Sell TY Friday after Mem. Day	Mem. Day + 1	Mem. Day + 4
Independence Day (July 4th)	Rotates	Long TY	After	Buy TY Independence Day + 2, Sell TY Independence Day + 4	Ind. Day + 2	Ind. Day + 4
Labor Day	Monday	Long TY	Before	Buy TY Monday before Labor Day, Sell TY Thursday before Labor Day	Labor Day - 4	Labor Day - 2
Columbus Day	Monday	Short TY	Before	Sell TY Monday before Col. Day, Buy TY Thursday before Col. Day	Colum. Day - 5	Colum. Day - 2
Thanksgiving	Thursday	Long TY	Before	Buy TY Friday before Thanksgiving, Sell TY Tuesday before Thanksgiving	Thanksgiving - 4	Thanksgiving - 2
Christmas	Rotates	Short TY	Before	Sell TY Christmas Day - 4, Buy TY Christmas Day - 1	Christmas - 4	Christmas - 1

Source: Morgan Stanley Research. Note: Past performance is no guarantee of future results.

Global Macro Strategy Team

MORGAN STANLEY & CO. LLC	**Matthew Hornbach, CMT** Matthew.Hornbach@morganstanley.com	Global Head of Macro Strategy	+1 212 761-1837
	Bradley Tian	Global Macro Commentary	+1 212 761-0826
	Martin Tobias, CFA, CMT	US Rates Strategist	+1 212 761-6076
	Shaun Zhou	US Rates Strategist	+1 212 761-3348
	Aryaman Singh	US Rates Strategist	+1 212 761-1993
	Eli Carter	US Rates Strategist	+1 212 761-4703
	Andrew Watrous	G10 FX Strategist	+1 212 761-5287
	Zoe K. Strauss	G10 FX Strategist	+1 212 761-0407
	Molly Nickolin	G10 FX Strategist	+1 212 761-3592
	Simon Waever Simon.Waever@morganstanley.com	Head of EM Sovereign Credit and Latin America Fixed Income Strategy	+1 212 296-8101
	Ioana Zamfir	Latin America Macro Strategist	+1 212 761-4012
	Emma Cerda	Latin America Sovereign Credit	+1 212 761-2344
	Sofia Palacios	Latin America Macro Strategist	+1 212 761-0428
MORGAN STANLEY & CO. INTERNATIONAL PLC	**James K. Lord** James.Lord@morganstanley.com	Global Head of FXEM Strategy	+44 20 7677-3254
	Gianluca Salford Luca.Salford@morganstanley.com	Head of European Rates Strategy	+44 20 7677-1337
	Lorenzo Testa	Euro Area Rates Strategist	+44 20 7677-0337
	Fabio Bassanin, CFA	UK Rates Strategist	+44 20 7425-1869
	Maria Chiara Russo	Euro Area Rates Strategist	+44 20 7677-3499
	David S. Adams, CFA David.S.Adams@morganstanley.com	Head of G10 FX Strategy	+44 20 7425-3518
	Neville Mandimika	CEEMEA Sovereign Credit Strategist	+44 20 7425-2509
	Arnav Gupta	CEEMEA Rates Strategist	+44 20 7677-0382
	David Cueva Oemer	CEEMEA Rates Strategist	+44 20 7425-2143
MORGAN STANLEY ASIA LIMITED+	**Gek Teng Khoo**	AXJ Macro Strategist	+852 3963-0303
MORGAN STANLEY MUFG SECURITIES CO., LTD.	**Koichi Sugisaki** Koichi.Sugisaki@morganstanleymufg.com	Head of Japan Macro Strategy	+81 3 6836-8428
	Hiromu Uezato	Japan Macro Strategist	+81 3 6836-8431

Disclosure Section

The information and opinions in Morgan Stanley Research were prepared by Morgan Stanley & Co. LLC, and/or Morgan Stanley C.T.V.M. S.A., and/or Morgan Stanley Mexico, Casa de Bolsa, S.A. de C.V., and/or Morgan Stanley Canada Limited. As used in this disclosure section, "Morgan Stanley" includes Morgan Stanley & Co. LLC, Morgan Stanley C.T.V.M. S.A., Morgan Stanley Mexico, Casa de Bolsa, S.A. de C.V., Morgan Stanley Canada Limited and their affiliates as necessary.

For important disclosures, stock price charts and equity rating histories regarding companies that are the subject of this report, please see the Morgan Stanley Research Disclosure Website at www.morganstanley.com/researchdisclosures, or contact your investment representative or Morgan Stanley Research at 1585 Broadway, (Attention: Research Management), New York, NY, 10036 USA.

For valuation methodology and risks associated with any recommendation, rating or price target referenced in this research report, please contact the Client Support Team as follows: US/Canada +1 800 303-2495; Hong Kong +852 2848-5999; Latin America +1 718 754-5444 (U.S.); London +44 (0)20-7425-8169; Singapore +65 6834-6860; Sydney +61 (0)2-9770-1505; Tokyo +81 (0)3-6836-9000. Alternatively you may contact your investment representative or Morgan Stanley Research at 1585 Broadway, (Attention: Research Management), New York, NY 10036 USA.

Analyst Certification

The following analysts hereby certify that their views about the companies and their securities discussed in this report are accurately expressed and that they have not received and will not receive direct or indirect compensation in exchange for expressing specific recommendations or views in this report: David S. Adams, CFA; Fabio Bassanin, CFA; Eli P Carter; David Cueva; Arnav Gupta; Matthew Hornbach; Gek Teng Khoo; James K Lord; Molly Nickolin; Sofia Palacios; Nimish M Prabhune; Maria Chiara Russo; Luca Salford; Aryaman Singh; Koichi Sugisaki; Lorenzo Testa; Bradley Tian; Martin W Tobias, CFA; Hiromu Uezato; Andrew M Watrous; Ioana Zamfir; Shaun Zhou.

.

Global Research Conflict Management Policy

Morgan Stanley Research has been published in accordance with our conflict management policy, which is available at www.morganstanley.com/institutional/research/conflictpolicies. A Portuguese version of the policy can be found at www.morganstanley.com.br

Important Regulatory Disclosures on Subject Companies

Within the last 12 months, Morgan Stanley managed or co-managed a public offering (or 144A offering) of securities of Germany.

Within the last 12 months, Morgan Stanley has received compensation for investment banking services from Germany.

In the next 3 months, Morgan Stanley expects to receive or intends to seek compensation for investment banking services from Germany, Japan, United Kingdom, United States of America.

Within the last 12 months, Morgan Stanley has received compensation for products and services other than investment banking services from Germany.

Within the last 12 months, Morgan Stanley has provided or is providing investment banking services to, or has an investment banking client relationship with, the following company: Germany, Japan, United Kingdom, United States of America.

Within the last 12 months, Morgan Stanley has either provided or is providing non-investment banking, securities-related services to and/or in the past has entered into an agreement to provide services or has a client relationship with the following company: Germany.

The equity research analysts or strategists principally responsible for the preparation of Morgan Stanley Research have received compensation based upon various factors, including quality of research, investor client feedback, stock picking, competitive factors, firm revenues and overall investment banking revenues. Equity Research analysts' or strategists' compensation is not linked to investment banking or capital markets transactions performed by Morgan Stanley or the profitability or revenues of particular trading desks.

Morgan Stanley and its affiliates do business that relates to companies/instruments covered in Morgan Stanley Research, including market making, providing liquidity, fund management, commercial banking, extension of credit, investment services and investment banking. Morgan Stanley sells to and buys from customers the securities/instruments of companies covered in Morgan Stanley Research on a principal basis. Morgan Stanley may have a position in the debt of the Company or instruments discussed in this report. Morgan Stanley trades or may trade as principal in the debt securities (or in related derivatives) that are the subject of the debt research report.

Certain disclosures listed above are also for compliance with applicable regulations in non-US jurisdictions.

STOCK RATINGS

Morgan Stanley uses a relative rating system using terms such as Overweight, Equal-weight, Not-Rated or Underweight (see definitions below). Morgan Stanley does not assign ratings of Buy, Hold or Sell to the stocks we cover. Overweight, Equal-weight, Not-Rated and Underweight are not the equivalent of buy, hold and sell. Investors should carefully read the definitions of all ratings used in Morgan Stanley Research. In addition, since Morgan Stanley Research contains more complete information concerning the analyst's views, investors should carefully read Morgan Stanley Research, in its entirety, and not infer the contents from the rating alone. In any case, ratings (or research) should not be used or relied upon as investment advice. An investor's decision to buy or sell a stock should depend on individual circumstances (such as the investor's existing holdings) and other considerations.

Global Stock Ratings Distribution

(as of September 30, 2025)

The Stock Ratings described below apply to Morgan Stanley's Fundamental Equity Research and do not apply to Debt Research produced by the Firm.

For disclosure purposes only (in accordance with FINRA requirements), we include the category headings of Buy, Hold, and Sell alongside our ratings of Overweight, Equal-weight, Not-Rated and Underweight. Morgan Stanley does not assign ratings of Buy, Hold or Sell to the stocks we cover. Overweight, Equal-weight, Not-Rated and Underweight are not the equivalent of buy, hold, and sell but represent recommended relative weightings (see definitions below). To satisfy regulatory requirements, we correspond Overweight, our most positive stock rating, with a buy recommendation; we correspond Equal-weight and Not-Rated to hold and Underweight to sell recommendations, respectively.

	Coverage Universe		Investment Banking Clients (IBC)			Other Material Investment Services Clients (MISC)	
Stock Rating Category	Count	% of Total	Count	% of Total IBC	% of Rating Category	Count	% of Total Other MISC
Overweight/Buy	1499	41%	389	46%	26%	702	41%
Equal-weight/Hold	1618	44%	375	44%	23%	782	45%
Not-Rated/Hold	4	0%	1	0%	25%	1	0%
Underweight/Sell	577	16%	88	10%	15%	234	14%
Total	3,698		853			1719	

Data include common stock and ADRs currently assigned ratings. Investment Banking Clients are companies from whom Morgan Stanley received investment banking compensation in the last 12 months. Due to rounding off of decimals, the percentages provided in the "% of total" column may not add up to exactly 100 percent.

Analyst Stock Ratings

Overweight (O). The stock's total return is expected to exceed the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.

Equal-weight (E). The stock's total return is expected to be in line with the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.

Not-Rated (NR). Currently the analyst does not have adequate conviction about the stock's total return relative to the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.

Underweight (U). The stock's total return is expected to be below the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.

Unless otherwise specified, the time frame for price targets included in Morgan Stanley Research is 12 to 18 months.

Analyst Industry Views

Attractive (A): The analyst expects the performance of his or her industry coverage universe over the next 12-18 months to be attractive vs. the relevant broad market benchmark, as indicated below.

In-Line (I): The analyst expects the performance of his or her industry coverage universe over the next 12-18 months to be in line with the relevant broad market benchmark, as indicated below.

Cautious (C): The analyst views the performance of his or her industry coverage universe over the next 12-18 months with caution vs. the relevant broad market benchmark, as indicated below.

Benchmarks for each region are as follows: North America - S&P 500; Latin America - relevant MSCI country index or MSCI Latin America Index; Europe - MSCI Europe; Japan - TOPIX; Asia - relevant MSCI country index or MSCI sub-regional index or MSCI AC Asia Pacific ex Japan Index.

Important Disclosures for Morgan Stanley Smith Barney LLC Customers

Important disclosures regarding the relationship between the companies that are the subject of Morgan Stanley Research and Morgan Stanley Smith Barney LLC or Morgan Stanley or any of their affiliates, are available on the Morgan Stanley Wealth Management disclosure website at www.morganstanley.com/online/researchdisclosures. For Morgan Stanley specific disclosures, you may refer to www.morganstanley.com/researchdisclosures.

Each Morgan Stanley research report is reviewed and approved on behalf of Morgan Stanley Smith Barney LLC. This review and approval is conducted by the same person who reviews the research report on behalf of Morgan Stanley. This could create a conflict of interest.

Other Important Disclosures

Morgan Stanley & Co. International PLC and its affiliates have a significant financial interest in the debt securities of Germany, Japan, United Kingdom.

Morgan Stanley Research policy is to update research reports as and when the Research Analyst and Research Management deem appropriate, based on developments with the issuer, the sector, or the market that may have a material impact on the research views or opinions stated therein. In addition, certain Research publications are intended to be updated on a regular periodic basis (weekly/monthly/quarterly/annual) and will ordinarily be updated with that frequency, unless the Research Analyst and Research Management determine that a different publication schedule is appropriate based on current conditions.

Morgan Stanley is not acting as a municipal advisor and the opinions or views contained herein are not intended to be, and do not constitute, advice within the meaning of Section 975 of the Dodd-Frank Wall Street Reform and Consumer Protection Act.

Morgan Stanley produces an equity research product called a "Tactical Idea." Views contained in a "Tactical Idea" on a particular stock may be contrary to the recommendations or views expressed in research on the same stock. This may be the result of differing time horizons, methodologies, market events, or other factors. For all research available on a particular stock, please contact your sales representative or go to Matrix at http://www.morganstanley.com/matrix.

Morgan Stanley Research is provided to our clients through our proprietary research portal on Matrix and also distributed electronically by Morgan Stanley to clients. Certain, but not all, Morgan Stanley Research products are also made available to clients through third-party vendors or redistributed to clients through alternate electronic means as a convenience. For access to all available Morgan Stanley Research, please contact your sales representative or go to Matrix at http://www.morganstanley.com/matrix.

Any access and/or use of Morgan Stanley Research is subject to Morgan Stanley's Terms of Use (http://www.morganstanley.com/terms.html). By accessing and/or using Morgan Stanley Research, you are indicating that you have read and agree to be bound by our Terms of Use (http://www.morganstanley.com/terms.html). In addition you consent to Morgan Stanley processing your personal data and using cookies in accordance with our Privacy Policy and our Global Cookies Policy (http://www.morganstanley.com/privacy_pledge.html), including for the purposes of setting your preferences and to collect readership data so that we can deliver better and more personalized service and products to you. To find out more information about how Morgan Stanley processes personal data, how we use cookies and how to reject cookies see our Privacy Policy and our Global Cookies Policy (http://www.morganstanley.com/privacy_pledge.html). Please use the provided link to review the Terms and Conditions and Most Important Terms and Conditions for Morgan Stanley India Company Private Limited (https://www.morganstanley.com/assets/pdfs/about-us-global-offices/india/Terms_and_conditions.pdf)

If you do not agree to our Terms of Use and/or if you do not wish to provide your consent to Morgan Stanley processing your personal data or using cookies please do not access our research.

Morgan Stanley Research does not provide individually tailored investment advice. Morgan Stanley Research has been prepared without regard to the circumstances and objectives of those who receive it. Morgan Stanley recommends that investors independently evaluate particular investments and strategies, and encourages investors to seek the advice of a financial adviser. The appropriateness of an investment or strategy will depend on an investor's circumstances and objectives. The securities, instruments, or strategies discussed in Morgan Stanley Research may not be suitable for all investors, and certain investors may not be eligible to purchase or participate in some or all of them. Morgan Stanley Research is not an offer to buy or sell or the solicitation of an offer to buy or sell any security/instrument or to participate in any particular trading strategy. The value of and income from your investments may vary because of changes in interest rates, foreign exchange rates, default rates, prepayment rates, securities/instruments prices, market indexes, operational or financial conditions of companies or other factors. There may be time limitations on the exercise of options or other rights in securities/instruments transactions. Past performance is not necessarily a guide to future performance. Estimates of future performance are based on assumptions that may not be realized. If provided, and unless otherwise stated, the closing price on the cover page is that of the primary exchange for the subject company's securities/instruments.

The fixed income research analysts, strategists or economists principally responsible for the preparation of Morgan Stanley Research have received compensation based upon various factors, including quality, accuracy and value of research, firm profitability or revenues (which include fixed income trading and capital markets profitability or revenues), client feedback and competitive factors. Fixed Income Research analysts', strategists' or economists' compensation is not linked to investment banking or capital markets transactions performed by Morgan Stanley or the profitability or revenues of particular trading desks.

The "Important Regulatory Disclosures on Subject Companies" section in Morgan Stanley Research lists all companies mentioned where Morgan Stanley owns 1% or more of a class of common equity securities of the companies. For all other companies mentioned in Morgan Stanley Research, Morgan Stanley may have an investment of less than 1% in securities/instruments or derivatives of securities/instruments of companies and may trade them in ways different from those discussed in Morgan Stanley Research. Employees of Morgan Stanley not involved in the preparation of Morgan Stanley Research may have investments in securities/instruments or derivatives of securities/instruments of companies mentioned and may trade them in ways different from those discussed in Morgan Stanley Research. Derivatives may be issued by Morgan Stanley or associated persons.

With the exception of information regarding Morgan Stanley, Morgan Stanley Research is based on public information. Morgan Stanley makes every effort to use reliable, comprehensive information, but we make no representation that it is accurate or complete. We have no obligation to tell you when opinions or information in Morgan Stanley Research change apart from when we intend to discontinue equity research coverage of a subject company. Facts and views presented in Morgan Stanley Research have not been reviewed by, and may not reflect information known to, professionals in other Morgan Stanley business areas, including investment banking personnel.

Morgan Stanley Research personnel may participate in company events such as site visits and are generally prohibited from accepting payment by the company of associated expenses unless pre-approved by authorized members of Research management.

Morgan Stanley may make investment decisions that are inconsistent with the recommendations or views in this report.

To our readers based in Taiwan or trading in Taiwan securities/instruments: Information on securities/instruments that trade in Taiwan is distributed by Morgan Stanley Taiwan Limited ("MSTL"). Such information is for your reference only. The reader should independently evaluate the investment risks and is solely responsible for their investment decisions. Morgan Stanley Research may not be distributed to the public media or quoted or used by the public media without the express written consent of Morgan Stanley. Any non-customer reader within the scope of Article 7-1 of the Taiwan Stock Exchange Recommendation Regulations accessing and/or receiving Morgan Stanley Research is not permitted to provide Morgan Stanley Research to any third party (including but not limited to related parties, affiliated companies and any other third parties) or engage in any activities regarding Morgan Stanley Research which may create or give the appearance of creating a conflict of interest. Information on securities/instruments that do not trade in Taiwan is for informational purposes only and is not to be construed as a recommendation or a solicitation to trade in such securities/instruments. MSTL may not execute transactions for clients in these securities/instruments.

Morgan Stanley is not incorporated under PRC law and the research in relation to this report is conducted outside the PRC. Morgan Stanley Research does not constitute an offer to sell or the solicitation of an offer to buy any securities in the PRC. PRC investors shall have the relevant qualifications to invest in such securities and shall be responsible for obtaining all relevant approvals, licenses, verifications and/or registrations from the relevant governmental authorities themselves. Neither this report nor any part of it is intended as, or shall constitute, provision of any consultancy or advisory service of securities investment as defined under PRC law. Such information is provided for your reference only.

Morgan Stanley Research is disseminated in Brazil by Morgan Stanley C.T.V.M. S.A. located at Av. Brigadeiro Faria Lima, 3600, 6th floor, São Paulo - SP, Brazil; and is regulated by the Comissão de Valores Mobiliários; in Mexico by Morgan Stanley México, Casa de Bolsa, S.A. de C.V which is regulated by Comision Nacional Bancaria y de Valores. Paseo de los Tamarindos 90, Torre 1, Col. Bosques de las Lomas Floor 29, 05120 Mexico City; in Japan by Morgan Stanley MUFG Securities Co., Ltd. and, for Commodities related research reports only, Morgan Stanley Capital Group Japan Co., Ltd; in Hong Kong by Morgan Stanley Asia Limited (which accepts responsibility for its contents) and by Morgan Stanley Bank Asia Limited; in Singapore by Morgan Stanley Asia (Singapore) Pte. (Registration number 199206298Z) and/or Morgan Stanley Asia (Singapore) Securities Pte Ltd (Registration number 200008434H), regulated by the Monetary Authority of Singapore (which accepts legal responsibility for its contents and should be contacted with respect to any matters arising from, or in connection with, Morgan Stanley Research) and by Morgan Stanley Bank Asia Limited, Singapore Branch (Registration number T14FC0118J); in Australia to "wholesale clients" within the meaning of the Australian Corporations Act by Morgan Stanley Australia Limited A.B.N. 67 003 734 576, holder of Australian financial services license No. 233742, which accepts responsibility for its contents; in Australia to "wholesale clients" and "retail clients" within the meaning of the Australian Corporations Act by Morgan Stanley Wealth Management Australia Pty Ltd (A.B.N. 19 009 145 555, holder of Australian financial services license No. 240813, which accepts responsibility for its contents; in Korea by Morgan Stanley & Co International plc, Seoul Branch; in India by Morgan Stanley India Company Private Limited having Corporate Identification No (CIN) U22990MH1998PTC115305, regulated by the Securities and Exchange Board of India ("SEBI") and holder of licenses as a Research Analyst (SEBI Registration No. INH000001105); Stock Broker (SEBI Stock Broker Registration No. INZ000244438), Merchant Banker (SEBI Registration No. INM000011203), and depository participant with National Securities Depository Limited (SEBI Registration No. IN-DP-NSDL-567-2021) having registered office at Altimus, Level 39 & 40, Pandurang Budhkar Marg, Worli, Mumbai 400018, India; Telephone no. +91-22-61181000; Compliance Officer Details: Mr. Tejarshi Hardas, Tel. No.: +91-22-61181000 or Email: tejarshi.hardas@morganstanley.com; Grievance officer details: Mr. Tejarshi Hardas, Tel. No.: +91-22-61181000 or Email: msic-compliance@morganstanley.com. Morgan Stanley India Company Private Limited (MSICPL) may use AI tools in providing research services. All recommendations contained herein are made by the duly qualified research analysts; in Canada by Morgan Stanley Canada Limited; in Germany and the European Economic Area where required by Morgan Stanley Europe S.E., authorised and regulated by Bundesanstalt fuer Finanzdienstleistungsaufsicht (BaFin) under the reference number 149169; in the US by Morgan Stanley & Co. LLC, which accepts responsibility for its contents. Morgan Stanley & Co. International plc, authorized by the Prudential Regulation Authority and regulated by the Financial Conduct Authority and the Prudential Regulation Authority, disseminates in the UK research that it has prepared, and research which has been prepared by any of its affiliates, only to persons who (i) are investment professionals falling within Article 19(5) of the Financial Services and Markets Act 2000 (Financial Promotion) Order 2005 (as amended, the "Order"); (ii) are persons who are high net worth entities falling within Article 49(2)(a) to (d) of the Order; or (iii) are persons to whom an invitation or inducement to engage in investment activity (within the meaning of section 21 of the Financial Services and Markets Act 2000, as amended) may otherwise lawfully be communicated or caused to be communicated. RMB Morgan Stanley Proprietary Limited is a member of the JSE Limited and A2X (Pty) Ltd. RMB Morgan Stanley Proprietary Limited is a joint venture owned equally by Morgan Stanley International Holdings Inc. and RMB Investment Advisory (Proprietary) Limited, which is wholly owned by FirstRand Limited. The information in Morgan Stanley Research is being disseminated by Morgan Stanley Saudi Arabia, regulated by the Capital Market Authority in the Kingdom of Saudi Arabia, and is directed at Sophisticated investors only.